ML-TEACHER PRD

Problem 1: Knowledge

Books
↓
Chunking
↓
Embeddings
↓
Vector DB
↓
Retriever

This is solved by RAG.

Problem 2: Teaching Style

Instruction
↓
Definition
↓
Math Intuition
↓
Code Example
↓
Interview Question
↓
Common Mistakes

This is solved by fine-tuning.

Combined Architecture

```
                    Books
                      ↓
                Docling/PyMuPDF
                      ↓
                  Chunking
                      ↓
                 Embeddings
                      ↓
                  ChromaDB
                      ↓
                  Retriever
                      ↓
 User Question → Fine-Tuned Small LLM
                      ↓
                   Answer
```

The RAG provides facts.

The fine-tuned model provides the teaching format.

About Gemma Generating the Dataset

You have two options.

Option A: Use Gemma Directly

```
Book Chunk
   ↓
Gemma
   ↓
Instruction-Response JSON
```

Example:

Input chunk:

Gradient Descent minimizes a cost function...

Output:

```
{
  "instruction":"Explain Gradient Descent",
  "response":"Definition...\nMath Intuition...\nCode Example..."
}
```

This is the fastest path.

Option B: Use Rules + Gemma

Instead of asking Gemma to invent everything:

Chunk
↓
Extract Topic
↓
Prompt Gemma
↓
Generate Structured Response

This tends to produce cleaner datasets.

For your first version, I'd just use Gemma.

Model Choice

For learning and local experimentation:

Tiny Models

- TinyLlama
- Qwen2.5-0.5B

Good:

Fast training
Low VRAM
Cheap

Bad:

Limited reasoning
Limited coding ability

Better Choice

Since you're teaching ML concepts:

- Qwen2.5-1.5B

This is probably the sweet spot.

Training Dataset Size

For behavior tuning:

100 examples → proof of concept
500 examples → noticeable improvement
1000+ examples → good consistency

You do not need 50,000 examples.

You are teaching format, not world knowledge.

What the Fine-Tuned Model Learns

Not:

Gradient Descent facts
Random Forest facts
PCA facts

RAG handles those.

It learns:

How to explain
How to structure
How to teach
How to present code

This is exactly the kind of task LoRA excels at.

Stage 1

Book
↓
Chunk
↓
Embedding
↓
ChromaDB
↓
RAG

Get retrieval working.

Stage 2

Book
↓
Gemma
↓
500–1000 JSON pairs
↓
LoRA Fine-Tune

Get the teaching style working.

Stage 3

RAG
 +
Fine-Tuned Model

Now you have both knowledge and teaching behavior.

Don't use a *too tiny* model just because it's small.**Qwen 2.5 1.5B** or **3B** for the tutor model. The extra quality is often worth it, especially for ML concepts where clarity matters.